P11. A Spatial-Domain Coordinated Control Method for CAVs at Unsignalized Intersections Considering Motion Uncertainty
中英双语博士精读笔记

Tong Zhao; Nikolce Murgovski; Baigen Cai; Wei ShangGuan

2025 · arXiv preprint

Theme / 主题： Spatial-domain robust trajectory planning
Method family / 方法族： Spatial-domain robust MPC / 空间域鲁棒模型预测控制
PDF pages / 页数： 15 **Relevance / 相关度：** 92/100
Source / 来源： https://arxiv.org/abs/2412.04290

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于"Spatial-domain robust trajectory planning"方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Spatial-domain robust trajectory planning. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点： Optimization-heavy; not a learned scheduler, and publication venue beyond arXiv not verified.

1.2 核心科学问题 / Core Scientific Question

在路径和速度存在不确定性时，如何把交叉口多类型冲突统一成可实时求解的空间域约束？

How can crossing, following, merging, and diverging conflicts be unified as real-time spatial-domain constraints under motion uncertainty?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution)：Transforms coordinated control into a spatial-domain nonlinear program with unified linear collision-avoidance constraints and real-time iteration.

- 安全/避碰贡献 (safety contribution): Handles crossing, following, merging, and diverging conflicts under path and speed uncertainty of HDVs.
- 平滑/停止避免贡献 (smoothing contribution): Generates smooth trajectories under state/control constraints, avoiding unnecessary braking where robust constraints permit.
- 创新力度评判 (reviewer judgement): 很强：空间域建模和统一线性碰撞约束对低层 smoother 安全可行性极有价值。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Spatial-domain robust MPC	空间域鲁棒模型预测控制	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言，确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading，对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验，判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式：空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints，并解释每个符号的物理意义？
- 能否画出输入-模型-算法-控制-验证的数据流图？
- 能否指出至少两个强假设，并说明它们在真实交通中何时失效？
- 能否复述实验基准是否公平，指标是否覆盖效率、安全、能耗和计算时间？
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验？

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

1

A Spatial-Domain Coordinated Control Method for CAVs at Unsignalized Intersections Considering Motion Uncertainty

Tong Zhao, Nikolce Murgovski, Baigen Cai, *Senior Member, IEEE*, and Wei ShangGuan, *Member, IEEE*

arXiv:2412.04290v2 [eess.SY] 23 Jun 2025

***Abstract*—Coordinated control of connected and automated vehicles (CAVs) emerges as a promising technology to improve traffic safety, efficiency, and sustainability. Meanwhile, mixed traffic, where CAVs coexist with conventional human-driven vehicles (HDVs), represents an upcoming and necessary stage in the development of intelligent transportation systems. Considering the motion uncertainty of HDVs, this paper proposes a coordinated control method for trajectory planning of CAVs at an unsignalized intersection in mixed traffic. By sampling in distance and using an exact change of variables, the coordinated control problem is formulated in the spatial domain as a nonlinear program, thereby allowing for unified linear collision avoidance constraints to handle vehicle crossing, following, merging, and diverging conflicts. The motion uncertainty of HDVs is decoupled and modeled as path uncertainty and speed uncertainty, whereby the robustness of collision avoidance is ensured in both spatial and temporal dimensions. The prediction deviation for HDVs is compensated by receding horizon optimization, and a real-time iteration (RTI) scheme is developed to improve computational efficiency. Simulation case studies are conducted to validate the efficacy, robustness, and potential for real-time application of the proposed methods. The results show that the proposed control scheme provides collision-free and smooth trajectories with state and control constraints satisfied. Compared with the converged baseline, the RTI scheme reduces the computation time by orders of magnitude, and the solution deviation is less than 2.3%, demonstrating a favorable trade-off between computational effort and optimality.**

***Index Terms*—Connected and automated vehicles (CAVs), coordinated control, trajectory planning, nonlinear optimization, model predictive control (MPC), unsignalized intersection.**

I. Introduction

A. Motivation

CONNECTED and automated vehicles (CAVs) are considered one of the disruptive technologies that are expected to change the transportation landscape [1], [2]. With the aid of vehicle-to-everything (V2X) communication, CAVs are able to achieve cooperative driving, which holds great potential for improving traffic safety, efficiency, and sustainability [3], [4].

Unsignalized intersections, being common and prone to crashes and congestion [5], [6], have been widely regarded as typical application scenarios for cooperative driving [7]. Various methods and techniques have been proposed for cooperative decision-making and control of CAVs at unsignalized intersections [8], [9], [10], [11]. However, most of them only work for fully cooperative systems with 100 % CAV penetration. Over a long period of time, CAVs and conventional human-driven vehicles (HDVs) will coexist and thus constitute mixed traffic, which is an inevitable stage in the evolution of intelligent transportation systems towards full autonomy. Therefore, it is significant to consider the motion uncertainty of HDVs and investigate coordinated control methods for CAVs applicable to unsignalized intersections in mixed traffic.

B. Literature Review

From the perspective of research ideas, existing approaches to cooperative driving at unsignalized intersections can be classified into two categories: 1) one-stage approaches, which directly solve for the trajectories of CAVs; 2) two-stage approaches, which decouple the cooperative driving problem into two parts: crossing scheduling and trajectory planning, and first schedule the crossing order of all vehicles, and then plan the trajectories of CAVs based on this. The one-stage approach tends to embed complex safety constraints, resulting in a highly nonlinear and nonconvex optimization problem that is difficult to solve [12], [13], [14]. In contrast, the two-stage approach reduces the problem complexity through decoupling and provides the flexibility that each stage can be handled by different methods. In the following, we focus on the research progress of the two-stage approach.

In terms of crossing scheduling, some strategies determine the crossing order based on heuristic rules, e.g., resource reservation [15], priority assignment [16], clearing policy [17]. Such schemes can be implemented quickly online, but their solutions are not guaranteed to be optimal or good enough. Another class of strategies solves for the optimal crossing order by constructing an optimization problem, typically a mixed-integer linear program [18], [19]. However, the computational complexity increases dramatically with the problem size, making these strategies intractable for real-time applications. To balance the computational effort and the optimality,

This work was supported in part by the Beijing Outstanding Young Scientist Program under Grant JWZQ20240101010; in part by the Joint Fund of the Ministry of Education for Equipment Pre-Research under Grant 8091B022238; and in part by the China Scholarship Council under Grant 202307090098. *(Corresponding author: Wei ShangGuan.)*

Tong Zhao is with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing 100044, China (e-mail: tong_zhao@bjtu.edu.cn).

Nikolce Murgovski is with the Department of Electrical Engineering, Chalmers University of Technology, 412 96 Gothenburg, Sweden (e-mail: nikolce.murgovski@chalmers.se).

Baigen Cai and Wei ShangGuan are with the School of Automation and Intelligence, the State Key Laboratory of Advanced Rail Autonomous Operation, and the Research Center for Autonomous Intelligence and Unmanned Systems, Beijing Jiaotong University, Beijing 100044, China (e-mail: bgcai@bjtu.edu.cn; wshg@bjtu.edu.cn).

图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接：Very strong for the smoother’s feasibility and robust collision-avoidance constraints.

II. Problem Formulation

This paper focuses on the coordinated control of CAVs at an unsignalized intersection in mixed traffic, i.e., traffic with both CAVs and HDVs. In this section, we introduce the intersection scenario, model vehicle kinematics, and formulate the coordinated control problem.

A. Unsignalized Intersection Scenario

Mathematically, any unsignalized intersection can be represented by the tuple

$$\left(p, x_l(p), y_l(p), \kappa_l(p), \psi_l(p), v_l^{\lim}(p), \mathcal{L}\right), \tag{1}$$

where p is a sampling variable denoting the distance traveled along the path, $x_l(p)$, $y_l(p)$ are the global coordinates of the predefined path $l \in \mathcal{L}$ at sample p, and $\kappa_l(p)$, $\psi_l(p)$, and $v_l^{\lim}(p)$ are the path curvature, tangent angle, and speed limit, respectively.

As a typical example, consider the four-way unsignalized intersection shown in Fig. 1, where vehicles are allowed to go straight, turn left, or turn right in the central area, depicted in gray. Each link consists of one entry lane and one exit lane, with the direction of travel shown by the arrows. Each entry lane leads to three paths to each exit lane that does not share a link with that entry lane, resulting in a total of 12 predefined paths. Finally, the red circle marks the control boundary, within which any vehicle is considered part of the intersection.

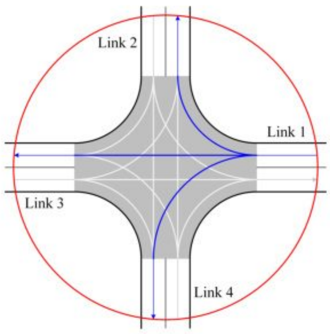


Fig. 1. An unsignalized intersection with four links, each with one entry lane and one exit lane. There are a total of 12 predefined paths, and the three paths starting from the first link are depicted in blue. The gray area is the central area. The red circle indicates the control boundary.

Now, consider $N \geq 1$ vehicles, consisting of M CAVs, with $1 \leq M \leq N$, and $N - M$ HDVs, traveling through the intersection, and assume that

- the crossing order has been given by a high-level scheduling module;
- the states (location, speed, etc.) of all vehicles can be captured at any time;
- the turning intentions of all vehicles are known and do not change;
- all CAVs are under centralized control;
- communication delays are negligible;
- lane changing is prohibited.

In practical terms, HDVs are sensed by CAVs and roadside units (RSUs), traffic information is shared through V2X communication, and centralized control is served by a certain CAV or RSU.

The set of all vehicle indices is denoted as $\mathcal{N} = \{1, \ldots, M, \ldots, N\}$, and its subset $\mathcal{M} = \{1, \ldots, M\}$ is defined as the set of CAV indices, and thus the set of HDV indices is $\mathcal{N} \setminus \mathcal{M}$. Note that the intersection illustrated in Fig. 1 is only an example used herein, and the methods herein can be applied to unsignalized intersections with different geometries.

B. Vehicle Paths

It is assumed that each vehicle $i \in \mathcal{N}$ is following a reference path

$$\boldsymbol{p}_{\mathrm{i}r}(p) = \begin{bmatrix} p & x_{\mathrm{i}r}(p) & y_{\mathrm{i}r}(p) & \kappa_{\mathrm{i}r}(p) & \psi_{\mathrm{i}r}(p) & v_{\mathrm{i}r}^{\lim}(p) \end{bmatrix}, \tag{2}$$

which is one of the predefined paths, i.e., $\boldsymbol{p}_{\mathrm{i}r}(p) = \begin{bmatrix} p & x_l(p) & y_l(p) & \kappa_l(p) & \psi_l(p) & v_l^{\lim}(p) \end{bmatrix}$ for some $l \in \mathcal{L}$. Thereby, the path of vehicle i can be expressed as

$$\begin{aligned} \boldsymbol{p}_i(p) &= \begin{bmatrix} p & x_i(p) & y_i(p) & \kappa_i(p) & \psi_i(p) & v_i^{\lim}(p) \end{bmatrix} \\ &= \boldsymbol{p}_{\mathrm{i}r}(\tilde{p}) + \boldsymbol{w}_i(\tilde{p}),\ \boldsymbol{w}_i(\tilde{p}) \in \mathscr{W}_i(\tilde{p}),\ \tilde{p} \in [\tilde{p}_{i0}, \tilde{p}_{i\mathrm{f}}], \end{aligned} \tag{3}$$

where $\tilde{p}$ is the projection of sample p onto the reference path, $\boldsymbol{w}_i(\tilde{p})$ is the path-following deviation of vehicle i and its range $\mathscr{W}_i(\tilde{p})$ is a complete bounded subset of $\mathbb{R}^6$, and $\tilde{p}_{i0}$ and $\tilde{p}_{i\mathrm{f}}$ are the projections of the initial position p_{i0} and the final position $p_{i\mathrm{f}}$ of vehicle i onto the reference path, respectively. Furthermore, CAVs are assumed to follow their respective reference paths perfectly, i.e., $\mathscr{W}_i(\tilde{p}) \equiv \{\mathbf{0}\}$ for all $i \in \mathcal{M}$. In contrast, the path-following deviations of HDVs are non-negligible and uncertain.

C. Vehicle Kinematics

For each vehicle $i \in \mathcal{N}$, let $\boldsymbol{x}_i(t) = \begin{bmatrix} p_i(t) & \dot{p}_i(t) \end{bmatrix}^\top$ denote its state vector, where t is the time, and $p_i(t)$ and $\dot{p}_i(t)$ are the longitudinal position and longitudinal velocity along its path, respectively. Then, vehicle i is represented by a linear system

$$\dot{\boldsymbol{x}}_i(t) = \mathbf{A}\boldsymbol{x}_i(t) + \mathbf{B}u_i(t), \tag{4}$$

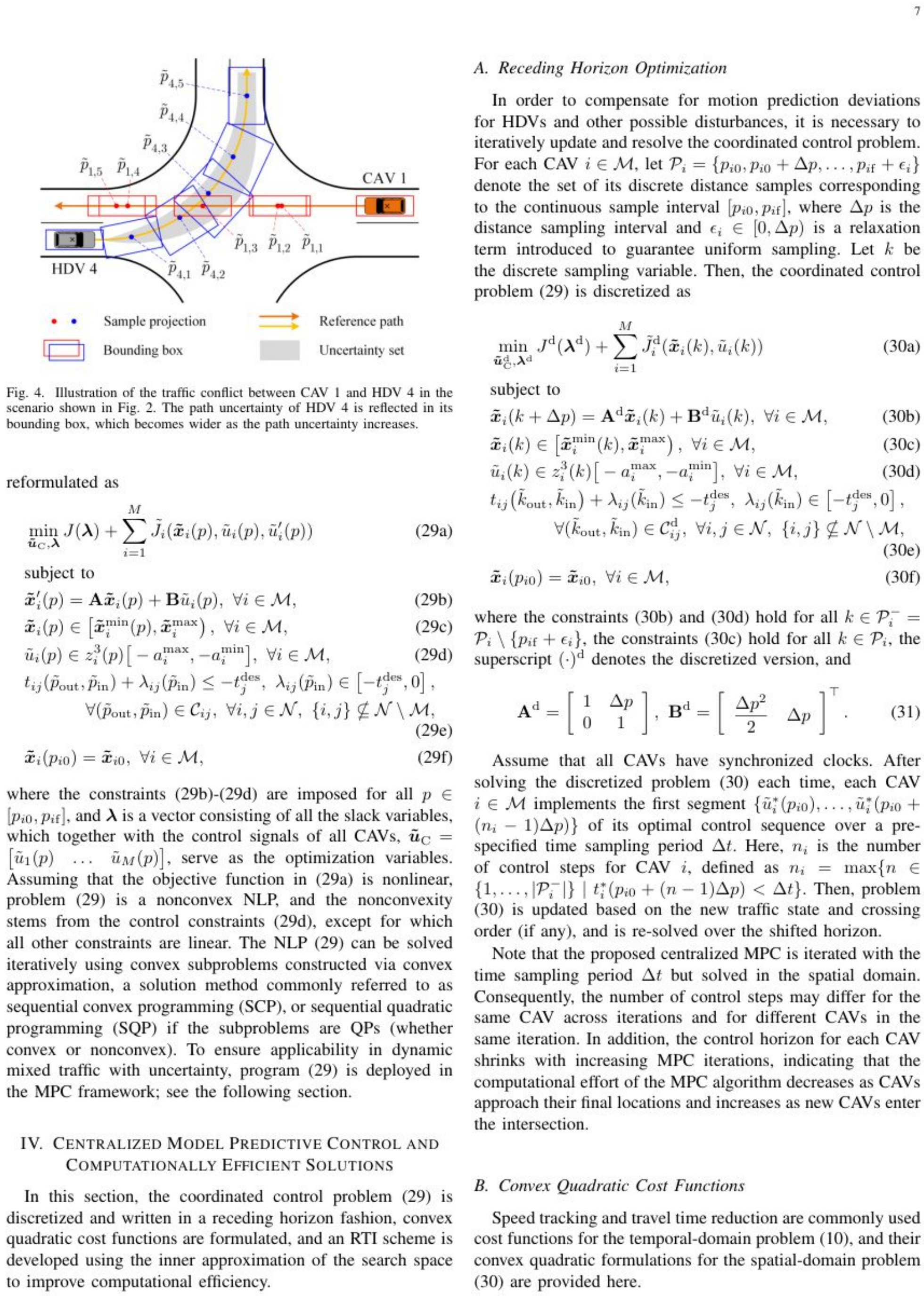
where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},\ \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tag{5}$$

and the longitudinal acceleration along its path serves as the control signal, i.e., $u_i(t) = \ddot{p}_i(t)$.

The predefined path, combined with the longitudinal acceleration and longitudinal velocity profiles along it, completely defines the trajectory of a CAV. This trajectory can be tracked in real-time based on fine-grained vehicle dynamics to generate actionable control commands, specifically drive torque, braking force, and steering angle. Nevertheless, trajectory tracking is beyond the scope of this paper, which focuses on trajectory planning.

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.3, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接：Very strong for the smoother’s feasibility and robust collision-avoidance constraints.



7

Fig. 4. Illustration of the traffic conflict between CAV 1 and HDV 4 in the scenario shown in Fig. 2. The path uncertainty of HDV 4 is reflected in its bounding box, which becomes wider as the path uncertainty increases.

reformulated as

$$\min_{\tilde{\boldsymbol{u}}_{\mathrm{C}},\boldsymbol{\lambda}} J(\boldsymbol{\lambda}) + \sum_{i=1}^{M} \tilde{J}_i(\tilde{\boldsymbol{x}}_i(p), \tilde{u}_i(p), \tilde{u}_i'(p)) \tag{29a}$$

subject to

$$\tilde{\boldsymbol{x}}_i'(p) = \mathbf{A}\tilde{\boldsymbol{x}}_i(p) + \mathbf{B}\tilde{u}_i(p),\ \forall i \in \mathcal{M}, \tag{29b}$$

$$\tilde{\boldsymbol{x}}_i(p) \in \left[\tilde{\boldsymbol{x}}_i^{\min}(p), \tilde{\boldsymbol{x}}_i^{\max}\right),\ \forall i \in \mathcal{M}, \tag{29c}$$

$$\tilde{u}_i(p) \in z_i^3(p)\left[-a_i^{\max}, -a_i^{\min}\right],\ \forall i \in \mathcal{M}, \tag{29d}$$

$$t_{ij}(\tilde{p}_{\mathrm{out}}, \tilde{p}_{\mathrm{in}}) + \lambda_{ij}(\tilde{p}_{\mathrm{in}}) \le -t_j^{\mathrm{des}},\ \lambda_{ij}(\tilde{p}_{\mathrm{in}}) \in \left[-t_j^{\mathrm{des}}, 0\right], \\ \forall(\tilde{p}_{\mathrm{out}}, \tilde{p}_{\mathrm{in}}) \in \mathcal{C}_{ij},\ \forall i,j \in \mathcal{N},\ \{i,j\} \not\subseteq \mathcal{N}\setminus\mathcal{M}, \tag{29e}$$

$$\tilde{\boldsymbol{x}}_i(p_{i0}) = \tilde{\boldsymbol{x}}_{i0},\ \forall i \in \mathcal{M}, \tag{29f}$$

where the constraints (29b)-(29d) are imposed for all $p \in [p_{i0}, p_{if}]$, and $\boldsymbol{\lambda}$ is a vector consisting of all the slack variables, which together with the control signals of all CAVs, $\tilde{\boldsymbol{u}}_{\mathrm{C}} = [\tilde{u}_1(p) \ \ \dots \ \ \tilde{u}_M(p)]$, serve as the optimization variables. Assuming that the objective function in (29a) is nonlinear, problem (29) is a nonconvex NLP, and the nonconvexity stems from the control constraints (29d), except for which all other constraints are linear. The NLP (29) can be solved iteratively using convex subproblems constructed via convex approximation, a solution method commonly referred to as sequential convex programming (SCP), or sequential quadratic programming (SQP) if the subproblems are QPs (whether convex or nonconvex). To ensure applicability in dynamic mixed traffic with uncertainty, program (29) is deployed in the MPC framework; see the following section.

IV. Centralized Model Predictive Control and Computationally Efficient Solutions

In this section, the coordinated control problem (29) is discretized and written in a receding horizon fashion, convex quadratic cost functions are formulated, and an RTI scheme is developed using the inner approximation of the search space to improve computational efficiency.

A. Receding Horizon Optimization

In order to compensate for motion prediction deviations for HDVs and other possible disturbances, it is necessary to iteratively update and resolve the coordinated control problem. For each CAV $i \in \mathcal{M}$, let $\mathcal{P}_i = \{p_{i0}, p_{i0} + \Delta p, \dots, p_{if} + \epsilon_i\}$ denote the set of its discrete distance samples corresponding to the continuous sample interval $[p_{i0}, p_{if}]$, where Δp is the distance sampling interval and $\epsilon_i \in [0, \Delta p)$ is a relaxation term introduced to guarantee uniform sampling. Let k be the discrete sampling variable. Then, the coordinated control problem (29) is discretized as

$$\min_{\tilde{\boldsymbol{u}}_{\mathrm{C}}^{\mathrm{d}},\boldsymbol{\lambda}^{\mathrm{d}}} J^{\mathrm{d}}(\boldsymbol{\lambda}^{\mathrm{d}}) + \sum_{i=1}^{M} \tilde{J}_i^{\mathrm{d}}(\tilde{\boldsymbol{x}}_i(k), \tilde{u}_i(k)) \tag{30a}$$

subject to

$$\tilde{\boldsymbol{x}}_i(k+\Delta p) = \mathbf{A}^{\mathrm{d}}\tilde{\boldsymbol{x}}_i(k) + \mathbf{B}^{\mathrm{d}}\tilde{u}_i(k),\ \forall i \in \mathcal{M}, \tag{30b}$$

$$\tilde{\boldsymbol{x}}_i(k) \in \left[\tilde{\boldsymbol{x}}_i^{\min}(k), \tilde{\boldsymbol{x}}_i^{\max}\right),\ \forall i \in \mathcal{M}, \tag{30c}$$

$$\tilde{u}_i(k) \in z_i^3(k)\left[-a_i^{\max}, -a_i^{\min}\right],\ \forall i \in \mathcal{M}, \tag{30d}$$

$$t_{ij}(\tilde{k}_{\mathrm{out}}, \tilde{k}_{\mathrm{in}}) + \lambda_{ij}(\tilde{k}_{\mathrm{in}}) \le -t_j^{\mathrm{des}},\ \lambda_{ij}(\tilde{k}_{\mathrm{in}}) \in \left[-t_j^{\mathrm{des}}, 0\right], \\ \forall(\tilde{k}_{\mathrm{out}}, \tilde{k}_{\mathrm{in}}) \in \mathcal{C}_{ij}^{\mathrm{d}},\ \forall i,j \in \mathcal{N},\ \{i,j\} \not\subseteq \mathcal{N}\setminus\mathcal{M}, \tag{30e}$$

$$\tilde{\boldsymbol{x}}_i(p_{i0}) = \tilde{\boldsymbol{x}}_{i0},\ \forall i \in \mathcal{M}, \tag{30f}$$

where the constraints (30b) and (30d) hold for all $k \in \mathcal{P}_i^- = \mathcal{P}_i \setminus \{p_{if} + \epsilon_i\}$, the constraints (30c) hold for all $k \in \mathcal{P}_i$, the superscript $(\cdot)^{\mathrm{d}}$ denotes the discretized version, and

$$\mathbf{A}^{\mathrm{d}} = \begin{bmatrix} 1 & \Delta p \\ 0 & 1 \end{bmatrix},\ \mathbf{B}^{\mathrm{d}} = \begin{bmatrix} \frac{\Delta p^2}{2} & \Delta p \end{bmatrix}^{\top}. \tag{31}$$

Assume that all CAVs have synchronized clocks. After solving the discretized problem (30) each time, each CAV $i \in \mathcal{M}$ implements the first segment $\{\tilde{u}_i^*(p_{i0}), \dots, \tilde{u}_i^*(p_{i0} + (n_i - 1)\Delta p)\}$ of its optimal control sequence over a pre-specified time sampling period Δt. Here, n_i is the number of control steps for CAV i, defined as $n_i = \max\{n \in \{1, \dots, |\mathcal{P}_i^-|\} \mid t_i^*(p_{i0} + (n-1)\Delta p) < \Delta t\}$. Then, problem (30) is updated based on the new traffic state and crossing order (if any), and is re-solved over the shifted horizon.

Note that the proposed centralized MPC is iterated with the time sampling period Δt but solved in the spatial domain. Consequently, the number of control steps may differ for the same CAV across iterations and for different CAVs in the same iteration. In addition, the control horizon for each CAV shrinks with increasing MPC iterations, indicating that the computational effort of the MPC algorithm decreases as CAVs approach their final locations and increases as new CAVs enter the intersection.

B. Convex Quadratic Cost Functions

Speed tracking and travel time reduction are commonly used cost functions for the temporal-domain problem (10), and their convex quadratic formulations for the spatial-domain problem (30) are provided here.

图 3: 核心方法或算法页 / Core method or algorithm page (p.7, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Very strong for the smoother's feasibility and robust collision-avoidance constraints.

1) Speed Tracking: The reference speed is tracked using the cost function

$$\tilde{J}_i^{\mathrm{d}}(\cdot) = \tilde{J}_{i1}^{\mathrm{d}}(\tilde{\boldsymbol{x}}_i(k), \tilde{u}_i(k)) + \tilde{J}_{i2}^{\mathrm{d}}(\tilde{\boldsymbol{x}}_i(k), \tilde{u}_i(k)), \quad (32)$$

where $\tilde{J}_{i1}^{\mathrm{d}}(\cdot)$ and $\tilde{J}_{i2}^{\mathrm{d}}(\cdot)$ represent the costs within and beyond the actual finite prediction horizon, respectively, thereby effectively mimicking an infinite horizon. Specifically,

$$\begin{aligned}\tilde{J}_{i1}^{\mathrm{d}}(\cdot) = {} & q_{i1}(z_i(p_{\mathrm{if}} + \epsilon_i) - 1/v_{i\mathrm{r}}(p_{\mathrm{if}} + \epsilon_i))^2 \\ & + \sum_{k \in \mathcal{P}_i^-} \big[q_{i1}(z_i(k) - 1/v_{i\mathrm{r}}(k))^2 + r_i \tilde{u}_i^2(k) \\ & + e_i(\tilde{u}_i(k) - \tilde{u}_i(k - \Delta p))^2\big], \end{aligned} \quad (33)$$

where $v_{i\mathrm{r}}(\cdot)$ is the reference speed and q_{i1}, r_i, and e_i are positive weights. Penalties for the control signal and its difference are included to mitigate sudden shifts in acceleration and jerk, thereby reducing discomfort and energy consumption. The cost beyond the actual finite prediction horizon (extending to infinity) is formulated as

$$\tilde{J}_{i2}^{\mathrm{d}}(\cdot) = \sum_{k \in \mathcal{P}_i^+} \left[q_{i1}(z_i(k) - 1/v_{i\mathrm{r,s}})^2 + r_i \tilde{u}_i^2(k)\right], \quad (34)$$

where $\mathcal{P}_i^+ = \{p_{\mathrm{if}} + \epsilon_i, p_{\mathrm{if}} + \epsilon_i + \Delta p, \ldots, \infty\}$ and $v_{i\mathrm{r,s}}$ is the steady state reference speed. This cost is introduced to stabilize the second state of the system, $z_i(k)$. In the standard linear-quadratic regulator fashion, it can be shown that

$$\tilde{J}_{i2}^{\mathrm{d}}(\cdot) = P_i(z_i(p_{\mathrm{if}}) - 1/v_{i\mathrm{r,s}})^2, \quad (35)$$

where P_i is the solution to the discrete algebraic Riccati equation, explicitly expressed as

$$P_i = \frac{q_{i1}}{2} + \sqrt{\left(\frac{q_{i1}}{2}\right)^2 + \frac{q_{i1} r_i}{\Delta p^2}}. \quad (36)$$

2) Travel Time Reduction: The final travel time is minimized using the cost function

$$\begin{aligned}\tilde{J}_i^{\mathrm{d}}(\cdot) = {} & q_{i2} t_i(p_{\mathrm{if}} + \epsilon_i) \\ & + \sum_{k \in \mathcal{P}_i^-} \left[r_i \tilde{u}_i^2(k) + e_i(\tilde{u}_i(k) - \tilde{u}_i(k - \Delta p))^2\right], \end{aligned} \quad (37)$$

where q_{i2} is a positive weight and the summation term is a penalty for discomfort and energy consumption. At the traffic control level, this cost is often used to maximize throughput.

3) Penalties for Slack Variables: The cost of exploiting the slack variables is formulated as

$$J^{\mathrm{d}}(\boldsymbol{\lambda}^{\mathrm{d}}) = N_{\mathrm{s}} q_{\mathrm{s}} \|\boldsymbol{\lambda}^{\mathrm{d}}\|^2, \quad (38)$$

where N_{s} is the number of the discretized slack variables and q_{s} is a large positive weight.

C. Real-Time Iteration

Although the NLP (30) can be solved with off-the-shelf solvers, the process might be time-consuming if the solver has to run until convergence. A computationally efficient solution is to stop the SCP/SQP scheme after only one iteration in each MPC update, which is known as RTI. In order to apply RTI, we need to ensure that the feasible solution to the first subproblem of the SCP/SQP scheme always yields a feasible solution to the original problem (even when the slack variables are not used).

Recalling the continuous control constraints (29d), it is found that the lower bound is convex and the upper bound is concave. Therefore, the linearization of the control constraints (29d) is an inner approximation to the original search space, which means that feasible points within the linearized bounds are also feasible in the original nonconvex set. Using a first-order Taylor expansion, the constraints (29d) are linearized around the lethargy $z_{i,\mathrm{lin}}(p)$, which is then discretized to yield

$$\tilde{u}_i(k) \in \left[\tilde{u}_i^{\min}(k, z_i(k)), \tilde{u}_i^{\max}(k, z_i(k))\right], \ \forall i \in \mathcal{M}, \quad (39)$$

where

$$\begin{aligned}\tilde{u}_i^{\min}(k, z_i(k)) &= a_i^{\max}(2z_{i,\mathrm{lin}}(k) - 3z_i(k))z_{i,\mathrm{lin}}^2(k), \\ \tilde{u}_i^{\max}(k, z_i(k)) &= a_i^{\min}(2z_{i,\mathrm{lin}}(k) - 3z_i(k))z_{i,\mathrm{lin}}^2(k).\end{aligned} \quad (40)$$

Now, by replacing the nonconvex constraints (30d) in the original problem (30) with the linear constraints (39), the target subproblem is obtained, which is a computationally efficient convex QP. This allows the application of RTI.

It should be noted that in the first MPC iteration for each CAV $i \in \mathcal{M}$, since there is no previous solution available, the inverse of the reference and maximum speeds are chosen to linearize the control constraint for the speed tracking cost and the travel time reduction cost, respectively. Thereafter, for both cost functions, the linearization is performed about the previous solution, shifted by n_i samples.

V. Simulation Results

In this section, the proposed MPC is validated in the scenarios shown in Fig. 5, where the lane width is $4\,\mathrm{m}$, the central area is $30\,\mathrm{m}$ across, and the control boundary is a circle with a radius of $90\,\mathrm{m}$, whose center coincides with the center of the central area. The path speed limit is $v_l^{\mathrm{lim}}(p) \equiv 50\,\mathrm{km/h}$ for all $l \in \mathcal{L}$. For each CAV $i \in \mathcal{M}$, the longitudinal acceleration limits are $a_i^{\min} = -3.5\,\mathrm{m/s^2}$ and $a_i^{\max} = 2\,\mathrm{m/s^2}$, the maximum acceptable centripetal acceleration is $a_{i\mathrm{c}}^{\max} = 2\,\mathrm{m/s^2}$, and the maximum speed is

$$v_i^{\max}(p) = \begin{cases} 50\,\mathrm{km/h}, & \text{if } \kappa_i(p) = 0, \\ 21.0\,\mathrm{km/h}, & \text{if } \kappa_i(p) = 0.0588, \\ 18.4\,\mathrm{km/h}, & \text{if } \kappa_i(p) = 0.0769, \end{cases} \quad (41)$$

where the curvature $\kappa_i(p) = 0$ corresponds to going straight, $\kappa_i(p) = 0.0588$ corresponds to turning left, and $\kappa_i(p) = 0.0769$ corresponds to turning right. The desired time gap is set to $t_j^{\mathrm{des}} = 1.1\,\mathrm{s}$ for all $j \in \mathcal{N}$. The distance sampling interval is chosen to be $\Delta p = 1\,\mathrm{m}$. The time sampling period is chosen to be $\Delta t = 0.5\,\mathrm{s}$.

The two cost functions presented in Section IV-B, speed tracking and travel time reduction, are used in the case study with the weights

$$q_{i1} = \frac{w_{i1}\Delta p}{z_{im}^3}, \ r_i = \frac{w_{i2}\Delta p}{z_{im}^5}, \ e_i = \frac{w_{i3}}{\Delta p z_{im}^7}, \ q_{i2} = 500, \quad (42)$$

where $w_{i1} = 1$, $w_{i2} = 1$, and $w_{i3} = 0.5$ are the weights corresponding to the temporal-domain problem (10), as detailed in [29], and z_{im} is the mean value of the linearization

图 4: 实验或结果页 / Experiments or results page (p.8, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接：Very strong for the smoother’s feasibility and robust collision-avoidance constraints.

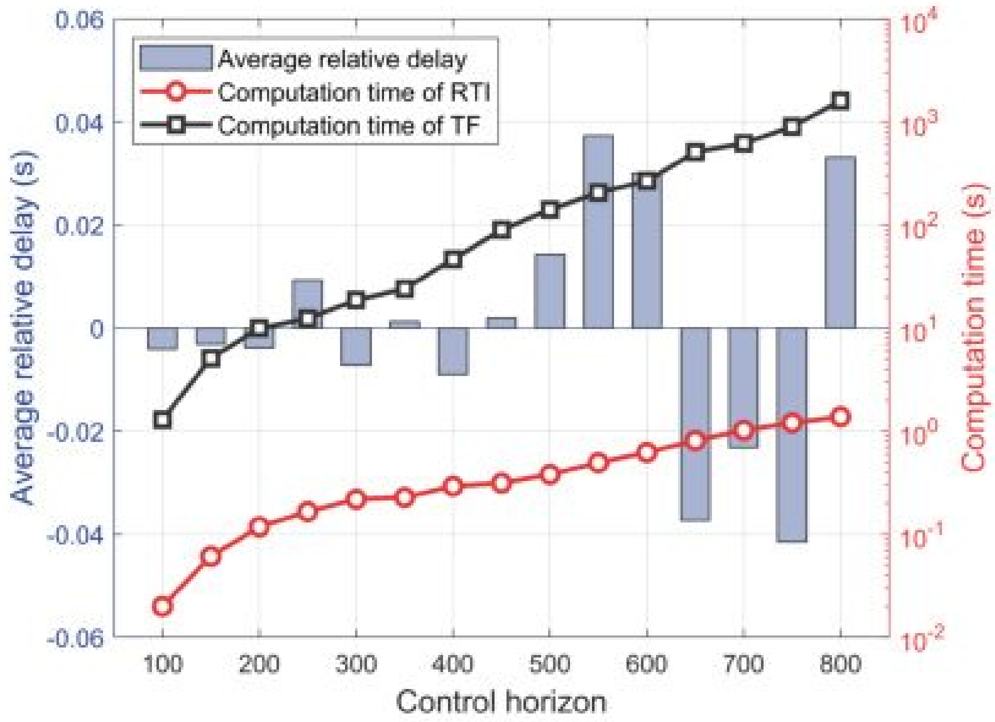


Fig. 15. Performance comparison between RTI and TF.

图 5: 结论或局限页 / Conclusion or limitations page (p.13, figure_crop). 这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接：Very strong for the smoother's feasibility and robust collision-avoidance constraints.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal：建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument：这一部分通常把痛点落到 Spatial-domain robust trajectory planning：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus：精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question：作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.3 II. PROBLEM FORMULATION

章节目标 / Learning goal：把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument：本节是复现论文的核心入口，应与公式“空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints”一起读。

技术焦点 / Technical focus：逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question：这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.4 III. M ODELING OF CONFLICT RESOLUTION CONSIDERING

章节目标 / Learning goal：承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument：这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus：检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question：跳过该节会不会导致公式或实验无法复现？

p.7 IV. C ENTRALIZED MODEL PREDICTIVE CONTROL AND

章节目标 / Learning goal：把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument：本节是复现论文的核心入口，应与公式“空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints”一起读。

技术焦点 / Technical focus：逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question：这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.8 V. SIMULATION RESULTS

章节目标 / Learning goal：验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument：实验应与论文声称的贡献闭环：Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.

技术焦点 / Technical focus：重点追踪指标：求解时间、最优性损失、碰撞约束余量、通行时间、速度平滑性和不确定性敏感性。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question：实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

p.10 6. The shaded space shows the motion uncertainty of HDV 4, where the X-Y

章节目标 / Learning goal：承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument：这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus：检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question：跳过该节会不会导致公式或实验无法复现？

p.10 1. Each CA V fulfills the speed and acceleration limits in all cases.

章节目标 / Learning goal：承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument：这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus：检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question：跳过该节会不会导致公式或实验无法复现？

p.13 VI. CONCLUSION AND FUTURE WORK

章节目标 / Learning goal：提炼作者承认的边界，并把它转化为博士课题的可拓展问题。

Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument：结论不只是总结结果，更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus：把 future work 与前文假设逐一对应，寻找可发表的补强点。

审稿追问 / Reviewer question：如果我是审稿人，我会要求补哪一个实验、定理或消融？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：在路径和速度存在不确定性时，如何把交叉口多类型冲突统一成可实时求解的空间域约束？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：车辆路径可用空间坐标参数化。；不确定性集合边界可估计且不严重失真。；集中式求解器可在控制周期内完成。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Transforms coordinated control into a spatial-domain nonlinear program with unified linear collision-avoidance constraints and real-time iteration.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Very strong for the smoother’s feasibility and robust collision-avoidance constraints.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints
3	核心求解 / Core solver	Transforms coordinated control into a spatial-domain nonlinear program with unified linear collision-avoidance constraints and real-time iteration.
4	安全接口 / Safety interface	Handles crossing, following, merging, and diverging conflicts under path and speed uncertainty of HDVs.
5	平滑与效率 / Smoothing and efficiency	Generates smooth trajectories under state/control constraints, avoiding unnecessary braking where robust constraints permit.
6	验证 / Validation	Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	4/5	空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints

算法贡献 / Algorithmic contribution	4/5	Transforms coordinated control into a spatial-domain nonlinear program with unified linear collision-avoidance constraints and real-time iteration.
实验贡献 / Experimental contribution	3/5	Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.
工程贡献 / Engineering contribution	4/5	Handles crossing, following, merging, and diverging conflicts under path and speed uncertainty of HDVs.
博士课题价值 / PhD relevance	5/5	Very strong for the smoother's feasibility and robust collision-avoidance constraints.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

空间域动力学与鲁棒碰撞约束 / Spatial-domain dynamics and robust collision constraints

$$\frac{dt_i}{ds} = \frac{1}{v_i(s)}, \quad \min_{u_i(s)} \sum_i \int \left(w_t \frac{1}{v_i(s)} + w_u u_i(s)^2 \right) ds, \quad h_{ij}(s, \Delta) \geq d_{safe}$$

Symbol / 符号	含义	Meaning	Role / 建模作用
s	空间路径坐标	spatial path coordinate	把时间规划转成沿路径规划
v_i(s)	空间域速度	speed over path coordinate	决定到达时间和安全间隔
h_ij	车辆 i,j 的冲突间隔函数	conflict-separation function	统一多种冲突类型
Delta	运动不确定性	motion uncertainty	路径/速度误差集合
d_safe	安全距离阈值	safe-distance threshold	鲁棒约束下界

5.2 核心算法架构 / Core Algorithm Layout

论文把轨迹规划从时间域转为空间域，针对 crossing/following/merging/diverging 构造统一线性化安全约束，再用集中式 MPC 和实时迭代求解，以较小最优性损失换取计算速度。

The method transforms coordination into a spatial-domain nonlinear program, linearizes unified collision-avoidance constraints, and solves it with real-time MPC iterations.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

鲁棒性来自显式不确定集合与安全约束，实时性来自空间域简化和 RTI。相比纯学习策略，这类 smoother 更容易给审稿人安全信心。

Robustness comes from explicit uncertainty sets and safety constraints; RTI makes the spatial-domain MPC viable online.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 车辆路径可用空间坐标参数化。
- 不确定性集合边界可估计且不严重失真。
- 集中式求解器可在控制周期内完成。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境： 时间域 MPC、非鲁棒控制、保守安全间隔策略和不同 RTI 近似。

Validation / 验证： Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.

Metrics / 指标： 求解时间、最优性损失、碰撞约束余量、通行时间、速度平滑性和不确定性敏感性。

Ablation / 消融： 很适合做鲁棒半径、RTI 次数、统一约束 vs 分类型约束的消融。

Reviewer reading / 审稿式阅读提醒： 阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 41 figure references and 5 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
控制框架/轨迹曲线图	control-framework or trajectory-profile plots	关注约束激活位置、速度/加速度曲线和安全裕度是否平滑。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 优化问题仍可能在大规模车辆数下变重。
- 不确定集合若估计过小会失去安全性，过大则过于保守。
- 高层排序/优先级选择不是主要贡献。

7.2 博士课题拓展点 / Research Extensions for PhD

- 将 JSSP 调度输出作为 MPC 的目标时间或优先级约束。
- 用学习模型预测不确定集合大小，实现自适应鲁棒性。
- 把空间域安全约束封装成调度动作可行性检查器。

7.3 如何服务当前课题 / Use in This Project

Very strong for the smoother's feasibility and robust collision-avoidance constraints.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter：不确定性半径 / Uncertainty radius, range [0.0, 3.0] m.
半径越大，安全裕度越高但轨迹越保守、延误越大。
A larger uncertainty radius improves safety margin but increases conservatism.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 <pre> 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.3 II. PROBLEM FORMULATION 把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。/ Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 III. M ODELING OF CONFLICT RESOLUTION CONSIDERING 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.7 IV. C ENTRALIZED MODEL PREDICTIVE CONTROL AND 把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。/ Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.8 V. SIMULATION RESULTS 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.10 6. The shaded space shows the motion uncertainty of HDV 4, where the X-Y 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.10 1. Each CA V fulfills the speed and acceleration limits in all cases. 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.13 VI. CONCLUSION AND FUTURE WORK 总结贡献和未来方向，也常暴露作者承认的边界条件。/ Summarizes contributions and often reveals author-recognized boundaries.